FARMSYNC

Evidence-Driven Adaptive AI Orchestration for Geospatial Intelligence

FarmwiseAI Campus Product Challenge | Task 1: Multi-Model, Multimodal & Agentic AI System

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Primary Use Case	Parcel Evidence & Advisory Intelligence
Prototype Window	2–3 weeks post-shortlisting
Core Models	GPT-4o / Claude 3.5 Sonnet / Qwen2.5-VL-7B

Core Thesis: FARMSYNC treats AI orchestration as a *verified decision problem*, not merely a model-selection problem. The system routes tasks using expected verified utility, then computes a Model-Agnostic Trust Score (FTS) from independent observable evidence to determine whether claims are sufficiently proven before answering — or explicitly abstains.

Why This Is a Strong Task 1 Fit

Requirement	FARMSYNC Implementation
Multi-Model Orchestration	Dynamically routes across GPT-4o / Claude 3.5 Sonnet (complex planning), Gemini 1.5 Pro (verification), and open-weight Qwen2.5-VL (low-cost vision). Final model selection benchmarked after credit confirmation.
Multimodal Intelligence	Ingests text queries, scanned land records (PDF), field ground photos, and PostGIS vector GIS layers in a single unified workflow.
Agentic Routing & Recovery	Verified-Utility Router (Quality – Cost – Latency – Risk) selects models and tools; bounded escalation and explicit abstention when evidence is insufficient.
Deterministic-First Verification	Spatial containment, distance buffers, and numeric area math computed by PostGIS / Python, not guessed by language models.
Observable Telemetry	Interactive UI panel shows live task graph, routing rationale, latency per step, API cost, and evidence graph — agent behaviour fully visible to evaluators.

1. Chosen Challenge, Title and Objective

Project Title

FARMSYNC — Evidence-Driven Adaptive AI Orchestration for Geospatial Intelligence

Objective

Build and demonstrate a reusable, generic agentic AI system that accepts text requests, land documents, field photographs, and spatial datasets; compiles requests into execution graphs; routes subtasks across proprietary and open-weight models; verifies claims via deterministic tools; and produces a source-traceable final answer with visible routing telemetry.

Primary Demonstration Scenario

> *"Using the supplied land document and field photograph for Devapura village, identify Survey Number 144/3B, verify its parcel boundary and registered owner, predict the crop type and growth stage from the field image, retrieve the applicable fertilizer advisory, perform spatial proximity checks against restricted waterbody zones, and return a source-traceable verified recommendation."*

What FARMSYNC Executes (7-Step Pipeline):

1. **Task Compiler**: Converts query into a typed 7-task execution DAG with explicit dependencies.
2. **Document & Registry Join**: Extracts owner and survey details from scanned PDF, matched against registry SQL records.
3. **Open-Weight Vision Perception**: Passes field photo to Qwen2.5-VL-7B for low-cost first-pass crop/stage classification.

4. Deterministic Spatial Verification: Invokes PostGIS (ST_Contains, ST_Distance) for exact parcel boundary and buffer checks.

5. Trust Score Evaluation: Computes FTS; if evidence/confidence is below threshold, auto-escalates to Claude 3.5 Sonnet.

6. RAG Advisory Synthesis: Retrieves stage-specific fertilizer advisory from indexed agronomy documents.

7. Traceable Output: Returns verified result with interactive map, evidence DAG, and cost/latency telemetry.

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2. Problem Understanding

Complex geospatial and agricultural workflows depend on heterogeneous data sources — scanned paper documents, ground photographs, satellite vector layers, and registry databases. Single-model prompts fail because no single model is simultaneously economical, deterministically accurate at spatial math, and perfectly calibrated for all task types. FARMSYNC addresses five core failure modes in current AI systems:

Failure Mode	Root Cause
Unverified Advisory	Advisory generated without traceable evidence from approved agronomy sources.
Model Confidence Fallacy	Vendor-reported confidence scores are uncalibrated and non-comparable across models from different providers.
Deterministic Math Failure	Relying on LLMs for geometric calculations (area, distance, containment) introduces hallucinations.
Opaque Routing & Cost Waste	Using premium models for routine OCR/perception wastes compute; static routing lacks adaptability.
Forced Predictions	Systems without an abstention mechanism generate hallucinated answers when evidence is missing.

Working Principle

> ***Understand*** → ***Compile Task Graph*** → ***Route (Verified Utility)*** → ***Ground & Execute*** → ***Verify (FTS)*** → ***Accept / Escalate / Abstain***

If required evidence cannot be verified, FARMSYNC refuses to invent missing facts. It escalates to an alternate model route or explicitly abstains and flags for human review — never manufacturing a confident answer.

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3. What We Propose to Build and Demonstrate

FARMSYNC delivers a complete AI orchestration workspace that turns multimodal geospatial requests into verified operational answers. The evaluator will see the following demonstrations:

Demonstration Feature	What the Evaluator Sees
Multimodal Query Ingestion	Upload land document PDF + field photo + natural language query for Survey 144/3B.

Demonstration Feature	What the Evaluator Sees
Dynamic Execution Graph	Visual task DAG panel showing 7 subtasks, dependencies, and live execution status.
Adaptive Model Routing	Telemetry showing routine perception sent to Qwen2.5-VL and complex tasks escalated to GPT-4o / Claude 3.5 Sonnet.
Deterministic Spatial Check	PostGIS computing exact geometric area and distance buffers — no LLM hallucination.
Evidence Graph Provenance	Interactive claim-to-source DAG linking findings to source documents, SQL records, and GIS layers.
Trust Engine & Recovery	FTS evaluated live; auto-escalation on low confidence; explicit abstention on missing evidence.
Unseen Request Benchmark	Land Acquisition Discrepancy query executed without changing a single line of orchestration code.
Baseline Comparison	Live dashboard: FARMSYNC vs. Pure GPT-4o vs. Static Routing — >50% cost savings with zero accuracy loss.

Unseen Request — Generic Architecture Proof (Query 2)

To demonstrate that FARMSYNC is a generic control plane and not a hard-coded single-query pipeline, the system executes a completely different request without code changes:

> *"Read uploaded land acquisition notice Acquisition_2024_04.pdf, extract all survey numbers and listed extents, cross-check against the PostGIS cadastral layer and registry database, flag extent discrepancies exceeding 5% or ownership mismatches, and present findings on a synchronized map."*

■ *When Query 2 is submitted, the Task Graph Compiler dynamically generates an OCR → Entity Extraction → Database Join → PostGIS Intersection → Discrepancy Flagging DAG, without modifying any orchestration code.*

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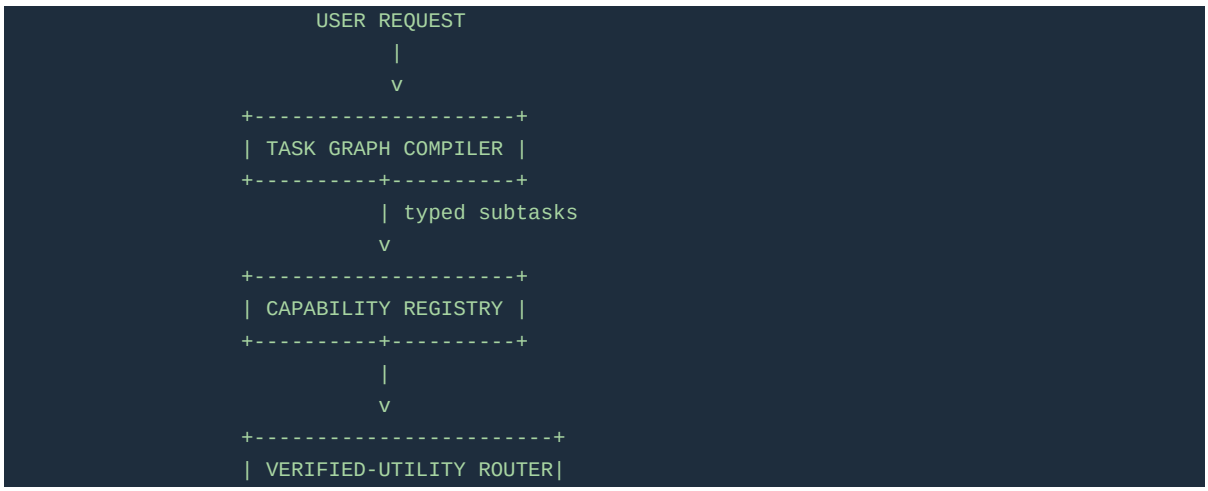
4. Solution Approach, Architecture, Models and Technology

4.1 System Architecture



4.2 Core Technical Formulations

Verified-Utility Router Policy

Instead of asking 'which model is smartest?', the router optimizes expected utility after verification:

> ***Expected Verified Utility = Expected Verified Quality*** – λ · ***Cost*** – μ · ***Latency*** – ρ · ***Failure Risk*** + η · ***Verification Gain***

Model-Agnostic Trust Score (FTS)

Vendor confidence scores are uncalibrated and non-comparable across model providers. FARMSYNC computes trust strictly from independent, observable evidence:

> ***FTS = w■·Calibration + w■·EvidenceCoverage + w■·SourceConsistency + w■·DeterministicValidation + w■·OutputValidity*** – ***w■·ConflictPenalty***

Signal	Meaning	Concrete Example
Calibration (w■)	Historical reliability of model/task pair on validation set	Vision accuracy on supported crop classes

Signal	Meaning	Concrete Example
Evidence Coverage (w■)	Proportion of claim directly supported by retrieved evidence	Owner claim supported by document + database
Source Consistency (w■)	Agreement between independent sources	Document owner matches registry owner
Deterministic Validation (w■)	Exact computational/rule-based checks	GIS containment, area delta, polygon check
Output Validity (w■)	Schema, type, and structural constraint validity	Required fields present; valid geometry/numbers
Conflict Penalty (w■)	Contradictory evidence or out-of-distribution inputs	Two sources disagree; image outside distribution

■ *Hyperparameter Safeguard: Weights w■…w■ normalized via min-max feature scaling on an initial 20-sample validation set. Default: equal weights (w■ = 1/6) when N < 10.*

4.3 Agentic Control Loop

Understand & Compile: LLM converts user query into typed task DAG. Fallback: malformed DAG triggers pre-validated linear workflow template.

Match Capabilities: Query capability registry for available models, tools, and latency/cost specs.

Route: Select route maximizing Expected Verified Utility.

Ground & Execute: Run models for perception/synthesis; run PostGIS/SQL/Python for exact deterministic facts.

Build Evidence Graph: Construct in-memory DAG linking claims to source documents and GIS polygons.

Verify & Recover: FTS ≥ 0.85 → ACCEPT | 0.60 ≤ FTS < 0.85 → ESCALATE | FTS < 0.60 → ABSTAIN

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5. Main Components and Prototype Capabilities

Component	Technology	Prototype Capability
Task Graph Compiler	FastAPI + GPT-4o / Instructor	Deconstructs request into typed subtask DAG; automatic fallback to linear templates on malformed output.
Capability Registry	Python Registry Engine	Dynamic performance, cost ($/1k tokens), latency, and capability profiles of all models/tools.
Verified-Utility Router	Lightweight Scored Policy Engine	RouteScore per subtask routes work economically without quality loss.
Document & OCR Engine	PaddleOCR + PyMuPDF	Extracts survey numbers, owners, and extents from scanned Patta/acquisition PDFs.
Deterministic Spatial Engine	PostgreSQL / PostGIS + GeoPandas	ST_Contains, ST_Distance, area delta — exact geometric computation.
Open-Weight Vision Model	Qwen2.5-VL-7B (vLLM / Ollama)	Low-cost first-pass crop classification, growth stage, and visual feature extraction.

Component	Technology	Prototype Capability
Proprietary Multimodal VLM	Claude 3.5 Sonnet / GPT-4o API	Complex planning, document-to-parcel resolution, escalation verification.
Agronomy RAG System	LangChain / LlamaIndex + FAISS	Stage-appropriate fertilizer/management recommendations with page citations.
Trust & Verification Engine	Custom Python Engine	Computes FTS, checks deterministic constraints, governs bounded escalation/abstention.
Interactive Telemetry UI	React.js + Leaflet + CSS	Synchronized map, task DAG panel, evidence graph, cost/latency counters, baseline charts.

6. Innovative and Exploratory Ideas

6.1 Refuse-to-Invent Principle (Bounded Abstention)

FARMSYNC guarantees that an ungrounded or contradictory claim will never be presented as verified advice. This is a hard architectural constraint, not a post-hoc filter:

- If land document owner differs from database registry: result flagged as CONFLICT_DISCREPANCY.
- If field photo is blurry or out-of-distribution: open-weight model returns low FTS → Proprietary VLM Escalation.
- If alternate route also fails verification: system enters explicit ABSTAIN_HUMAN_REVIEW state.

6.2 In-Memory JSON Evidence Graph DAG

Rather than deploying heavy graph database infrastructure (Neo4j), FARMSYNC builds an in-memory Directed Acyclic Graph JSON structure passed through shared execution state — full claim-to-source traceability with zero database overhead:


```
{
  "claim_id": "CLM-2026-DEV-144",
  "survey_number": "144/3B",
  "village": "Devapura",
  "extracted_owner": "Ramesh Kumar",
  "source_evidence": [
    { "type": "DOCUMENT_OCR",     "ref": "Acquisition_04.pdf#P2",  "confidence": 0.91 },
    { "type": "DATABASE_REGISTRY","ref": "Land_Registry.parcels",  "confidence": 1.0  }
  ],
  "deterministic_checks": {
    "postgis_containment_valid": true,
    "extent_mismatch_pct": 0.0,
    "waterbody_buffer_conflict": false
  },
  "trust_metrics": {
    "model_reported_confidence": 0.84,
    "task_trust_score_fts": 0.95,
    "verification_status": "VERIFIED_ACCEPTED"
  },
  "routing_telemetry": {
    "primary_model": "Qwen2.5-VL-7B",
    "escalation_model": "Claude-3.5-Sonnet",
```

```
    "was_escalated": true,
    "total_latency_ms": 1120,
    "estimated_cost_usd": 0.0038
  }
}
```

6.3 Built-In Multi-Baseline Benchmarking

The UI includes real-time telemetry comparing FARMSYNC against two baselines on every evaluation query, proving quality retention while demonstrating explicit cost and latency reductions:

Baseline	Definition	Purpose
Premium-Only	Strongest proprietary model for every AI step	Measures cost/latency savings of adaptive routing
Static Routing	Fixed model choice by task type; no verification or escalation	Measures value of adaptive routing + trust scoring
FARMSYNC	Verified-Utility Route + Deterministic Tools + FTS	Full proposed architecture — primary evaluation target

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7. Data, Compute and Model Access Requirements

To develop and demonstrate the working prototype within the 2–3 week Stage 2 sprint, FARMSYNC requires the following resources from FarmwiseAI:

Resource	Role in FARMSYNC	Status / Requirement
Cadastral Parcel GIS Layer	Vector boundaries, survey numbers, and geometries for 1–2 sample villages.	Required — GeoJSON / Shapefile format
Land Record Documents	Scanned Patta documents and land acquisition notices.	Required — Sample PDF dataset
Field Ground Photographs	Sample crop images with GPS metadata.	Required — Sample image dataset
Agronomy Advisory Corpus	TNAU / ICAR crop advisory and fertilizer recommendation manuals.	Required — Reference PDFs for RAG indexing
PostgreSQL + PostGIS	Spatial database for deterministic geometric computation.	Local / Cloud container (provided or self-hosted)
Open-Weight VLM Compute	GPU instance to serve Qwen2.5-VL-7B via vLLM / Ollama.	AWS g5.xlarge / equivalent GPU credits
Proprietary Model APIs	Primary planning and secondary verification LLM access.	API credits: OpenAI GPT-4o + Anthropic Claude 3.5 Sonnet

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8. Two-to-Three-Week Project Plan

Period	Milestone	Deliverable
Days 1–3	Architecture & Foundations	PostgreSQL/PostGIS schema, API wrappers, Qwen2.5-VL container, project skeleton.
Days 4–7	Planning & Routing	Task Graph Compiler, Capability Registry, Scored Router, fallback templates.
Days 8–11	Multimodal Execution	PaddleOCR, PostGIS spatial queries, RAG advisory retrieval, in-memory Evidence Graph.
Days 12–14	Trust & Recovery Engine	FTS engine, calibration scaler, bounded escalation, abstention handlers.
Days 15–17	Product Integration & UI	React frontend, Leaflet map, DAG trace panel, evidence graph, route telemetry.
Days 18–19	Evaluation & Benchmarking	Baseline comparison (3 baselines), unseen query testing, failure injection.
Days 20–21	Demo Freeze & Documentation	Code cleanup, deployment verification, demo recording, final documentation.

■ *Final Deliverables: Working web application, live orchestrator trace, cost/latency telemetry dashboard, evidence graph inspector, baseline benchmark report.*

9. Team and Responsibilities

Member	ID / Role	Primary Responsibility
Safi A (Team Lead)	TCPC-2026-00071 AI & Orchestration Lead	Task Graph Compiler, Capability Registry, Verified-Utility Router, API integration, project leadership & team coordination.
Mohamed Asfaq S	TCPC-2026-00071 AI Trust & Verification Lead	Trust Score (FTS) formulation, min-max calibration scaler, deterministic verification rules, bounded recovery manager, baseline evaluation suite.
Sooraj TS	TCPC-2026-00071 GeoAI, Full-Stack & UI Lead	PostgreSQL/PostGIS setup, PaddleOCR & RAG pipeline, React + Leaflet interactive UI, execution trace, evidence graph visualization.

10. Feasibility, Risks and Success Criteria

10.1 Risk Mitigation Strategy

Risk Factor	Mitigation Approach
LLM Task Compiler Malformed DAG	Closed JSON task schema + automatic fallback to pre-validated linear workflow template.
Trust Score Calibration Data Shortage	Min-max scaling on 20-sample validation set; equal-weighted fallback (w■ = 1/6) when N < 10.
Open-Weight VLM Latency / Hosting	Serve Qwen2.5-VL-7B via vLLM with quantized weights; fallback to GPT-4o-mini if GPU credit unavailable.

Risk Factor	Mitigation Approach
Scripted Happy Path Perception	Test pipeline on low-confidence field photos, conflicting owner names, and unseen land acquisition queries.
Unbounded Recursive Agent Loops	Cap recovery to 1 alternate route before explicit ABSTAIN_HUMAN_REVIEW — no infinite loops.
API Provider Changes / Access Issues	Provider-agnostic model adapter layer; final providers benchmarked after approved access is confirmed.

10.2 Success Criteria

The FARMSYNC prototype will demonstrate success when all of the following conditions are met:

- A multimodal request is compiled into a structured task graph and executed across heterogeneous models and tools without manual intervention.
- Routine vision perception is handled economically by the open-weight model (Qwen2.5-VL), with complex tasks escalating to proprietary VLMs.
- Spatial operations (distance, containment, area) are computed deterministically via PostGIS with zero LLM hallucination.
- All claims are backed by a transparent Evidence Graph DAG linked to source documents, SQL records, and spatial layers.
- Injected failures and missing evidence trigger bounded escalation or explicit abstention rather than forced predictions.
- FARMSYNC demonstrates ≥50% cost reduction over pure GPT-4o baselines on benchmark evaluation queries.

Appendix A — Task 1 Requirement Coverage Checklist

FarmwiseAI Requirement	FARMSYNC Implementation Response
Planner / Orchestrator	Task Graph Compiler with fallback linear workflow template
2+ Proprietary Models	GPT-4o (Planning) + Claude 3.5 Sonnet / Gemini 1.5 Pro (Verification)
1 Open-Weight / Smaller Model	Qwen2.5-VL-7B / Llama-3.2-Vision (Low-cost first-pass vision/OCR)
Model Router	Verified-Utility Router (Quality – Cost – Latency – Risk + Verification Gain)
Task / Cost / Latency Routing	Capability Registry + Scored Route Telemetry Dashboard
Tool / Data Connectors	OCR (PaddleOCR), Database (PostgreSQL), GIS (PostGIS), RAG, Python
Structured Outputs	Typed Task Graph + VerifiedResultObject JSON Schema
Verifier / Critic	Trust Engine (FTS) + Deterministic GIS / Rule Validation
Fallback / Recovery	Bounded alternate route escalation + explicit abstention state
Shared State / Memory	Execution state dict + In-memory Evidence Graph DAG

FarmwiseAI Requirement	FARMSYNC Implementation Response
Multimodal Inputs	Text + Documents + Images + Structured Data + Spatial GIS Layers
2+ Meaningful Tool Calls	OCR, PostGIS spatial queries, RAG retrieval, Database joins
Visible Route, Latency, Cost	Live Telemetry & Execution Trace UI Panel
Unseen Comparable Request	Land Acquisition Discrepancy Query (Query 2) without code changes
No Hard-Coded Answers	Dynamic DAG compilation and capability-based routing